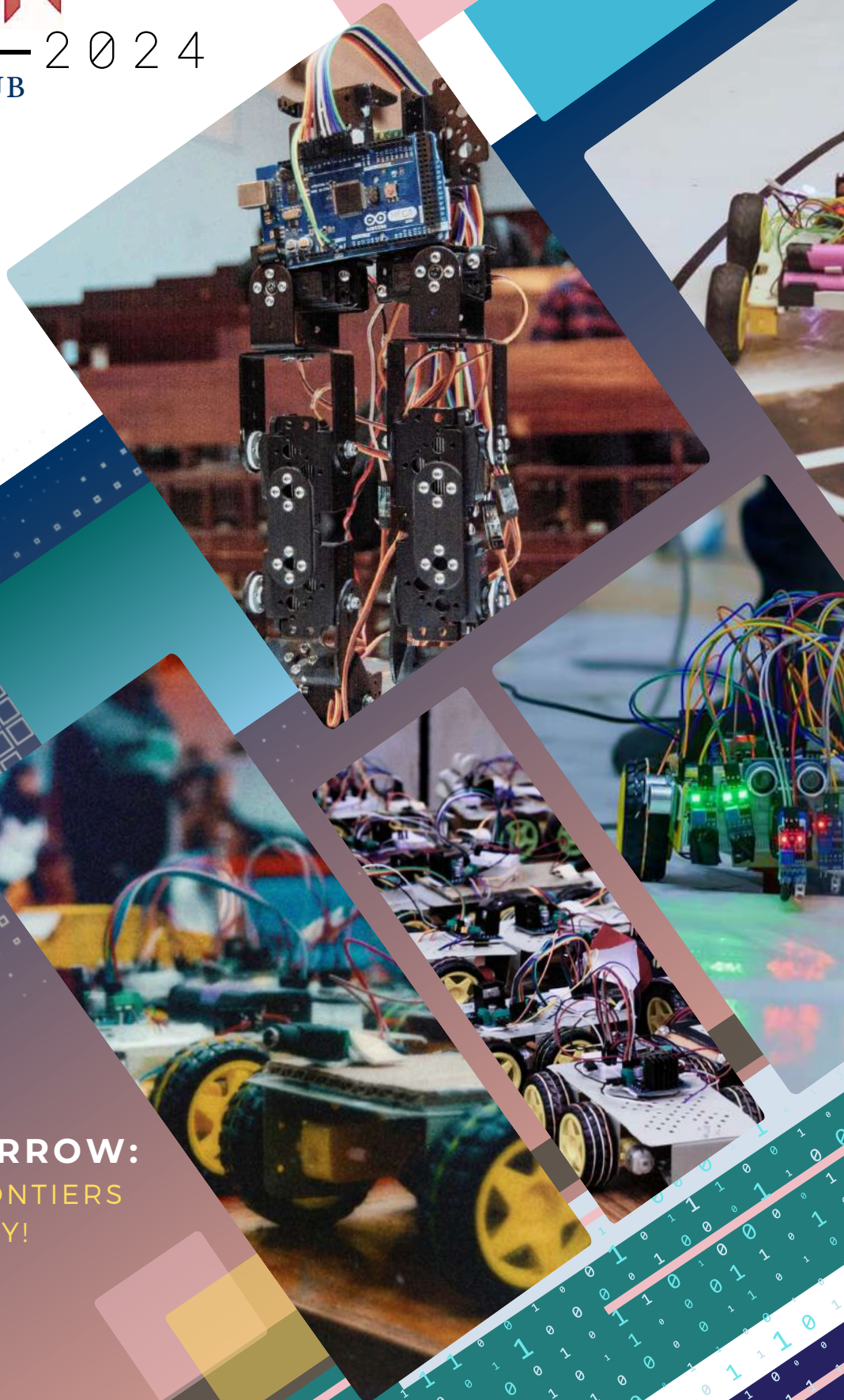
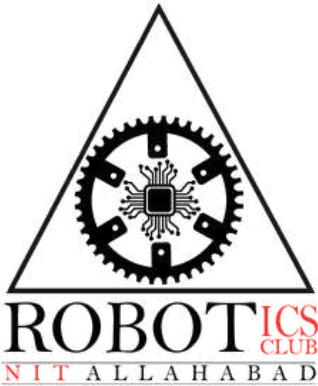




PRAGATI

2024

ANNUAL ROBOTICS CLUB
MAGAZINE



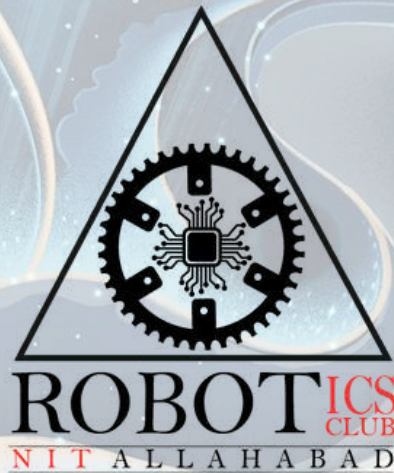
UNVEILING TOMORROW:
EXPLORING THE FRONTIERS
OF TECH TODAY!

PRAGATI

ANNUAL ROBOTICS CLUB MAGAZINE

ISSUE 01

A BIG LEAP TO FUTURE



MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY ALLAHABAD

PRAYAGRAJ - 211004

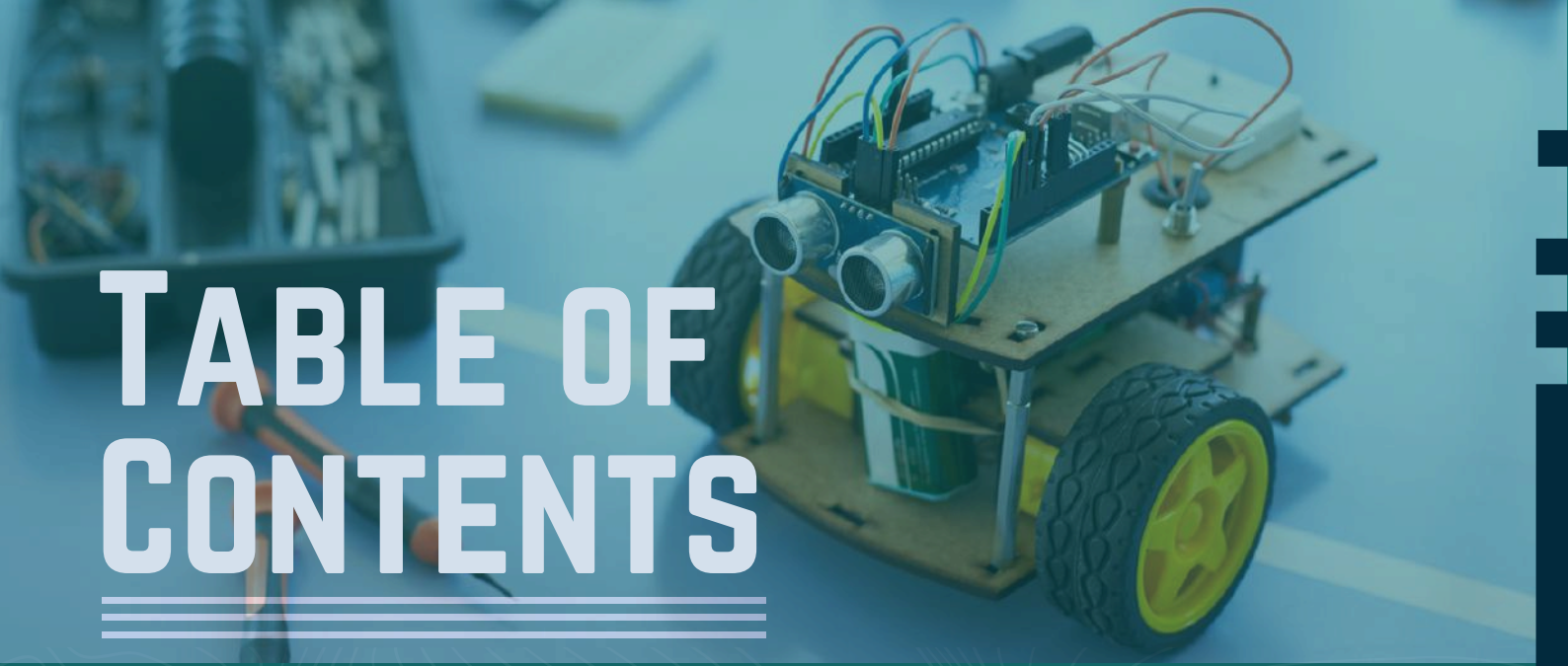


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INTRODUCTION

Welcome to the hub of innovation and creativity, where minds meet machines to redefine the future - Robotics Club MNNIT Allahabad. Nestled under the assistance of SAC MNNIT Allahabad, our club stands as an opportunity for students to embark on their journey of innovation and creativity.

Renowned for our boundless creativity, we foster a culture that nurtures innovation and a creative mindset among our members. From the inception of their college journey, we cater to students' thirst for knowledge and exploration.

Our hallmark lies in the diverse array of summer projects we unveil each year, designed to ignite the zeal for learning and innovation. These projects serve as a platform for students to push boundaries, explore new horizons, and turn ideas into reality.

We at Robotics Club organize events like Botrush and Robomania, drawing participation from over 2000 curious technocrats. These events serve as arenas to showcase their skills and maintain the zeal for innovation and positive competition among students.

Currently, we're steering towards the future with our renowned SDC(self-driving car) project, which has now advanced to phase two development. With each milestone, we edge closer to revolutionizing transportation through cutting-edge technology.

At Robotics Club MNNIT Allahabad, we believe in the phrase "**Innovate Automate Dominate**" and continuously strive for new ideas that can be turned into reality. Our journey is a testament to the power of creativity and the boundless possibilities it unlocks.

Let's look at different activities in the academic year 2023-24 under Robotics Club MNNIT Allahabad.





SUMMER PROJECTS

The robotics club MNNIT unleashes around 15+ Summer Projects ranging from various domains like Computer Vision, AI and ML, Inverse Kinematics, and Hardware Projects that are focused on solving real-life problems.

Prosthetic Arm

This project represents the first iteration of developing a prosthetic arm that can mimic the movements of a user's arm by leveraging flex sensors, an ESP32 microcontroller, and servo motors. The core components of this initial version included an ESP32 microcontroller board as the main controller, Flex sensors mounted on a glove to detect finger and wrist movements, Servo motors for actuating the movements of the prosthetic arm, 3D-printed prosthetic arm with basic joints and linkages. In this first version, the focus is on establishing the fundamental functionality and proof of concept. The flex sensors are integrated into a glove, and their data is wirelessly transmitted to the ESP32 microcontroller using its built-in Wi-Fi or Bluetooth capabilities.

The ESP32 is programmed to process the received data and convert it into corresponding commands for the servo motors. These servo motors are connected to a 3D-printed prosthetic arm with basic joints and linkages, allowing for simple arm movements to be replicated.

This first version serves as a foundation for future iterations, where more advanced features, such as improved ergonomics, better motion replication, and additional functionality, can be incorporated. The primary goal is to validate the concept and identify potential areas for improvement and refinement.



A Prosthetic Arm



A Prosthetic Arm





SUMMER PROJECTS

Humanoid

The project aims to develop a bipedal robot using Arduino which walks and moves like a human. The model consists of ten degrees of freedom, with each leg having five degrees of freedom. Both ends of the legs are connected to the torso. The Torso is a rigid body. This robot can move in a straight line in any direction using the concepts of inverse kinematics. At first, we acknowledged the different motions of our robot and through that motion, we planned the trajectories and found different positions of our robot. Then we used the concepts of inverse Kinematics to find different joint angles of the servo motor for the best possible walk. The inverse kinematics provide closed-form solutions to finding joint configurations that drive the end effectors of the robot to desired target positions in the three-dimensional physical space.



Team with Humanoid robot



5 DOF Robotic Arm

This project aims to develop a versatile and precise robotic arm with five degrees of freedom by using components such as servo motors and microcontrollers like Arduino. The goal is to create a robotic arm capable of performing complex tasks with accuracy and efficiency. The primary focus is on achieving accurate positioning and manipulation of objects in three-dimensional space. Through the implementation of forward and inverse kinematics, the robotic arm is capable of interpreting desired end-effector positions and generating corresponding joint configurations to achieve them. The ultimate goal is to create a versatile and reliable robotic platform suitable for tasks such as pick-and-place operations, assembly, and manipulation in industrial, research, and educational settings.



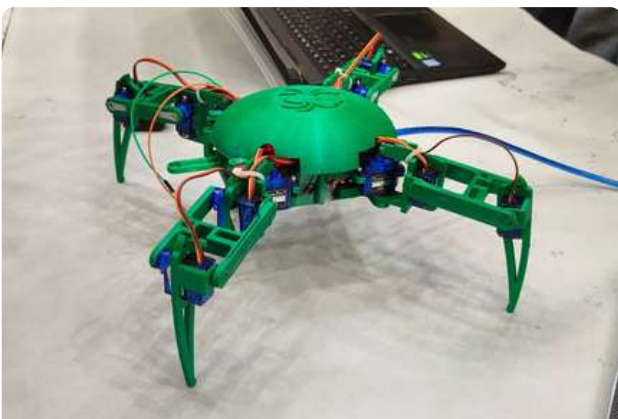


SUMMER PROJECTS

Spider Bot

This project aims to design and build a basic spider robot that can move and navigate on flat surfaces. The end goal of this project would be to create a functional and reliable spider robot that can move and navigate its environment. The Quadrupled Spider Robot is a four-legged walking robot inspired by the movement of spiders. It uses servo motors for each leg to achieve precise control and coordinated movement. The Arduino microcontroller is employed to control the servo motors, and the Adafruit_PWMServoDriver library is used for PWM signal generation.

The Spider Bot uses an Arduino Nano board as its microcontroller. An Adafruit PWM Servo Driver is employed to control the numerous SG90 Mini Servo Motors efficiently. Twelve(12) SG90 Mini Servo Motors are distributed across the four legs(3 per leg) and body of the Spider Bot. Inverse kinematics is employed to calculate the required joint angles for the robot to achieve specific positions or motions. Trajectory planning involves determining the time series of successive joint angles for coordinated leg and body movements. The Spider Bot can execute turns and shifts using a combination of trajectory planning and inverse kinematics. The Spider Bot executes a sequence of leg movements, including lifting, swinging, and placing, to achieve forward motion. Coordinated movements of all legs are orchestrated to mimic the walking cycle.



A Spider - Bot



A Spider - Bot



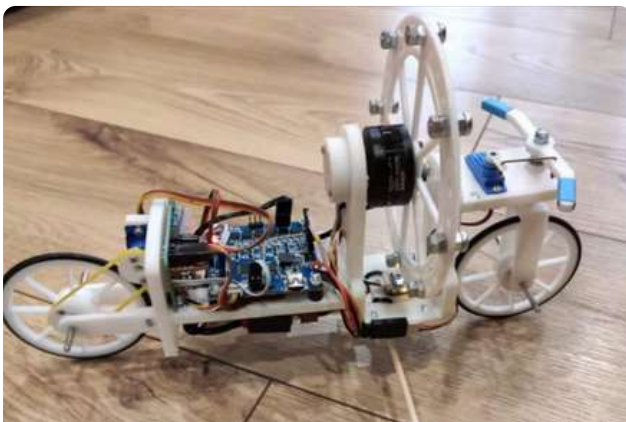


SUMMER PROJECTS

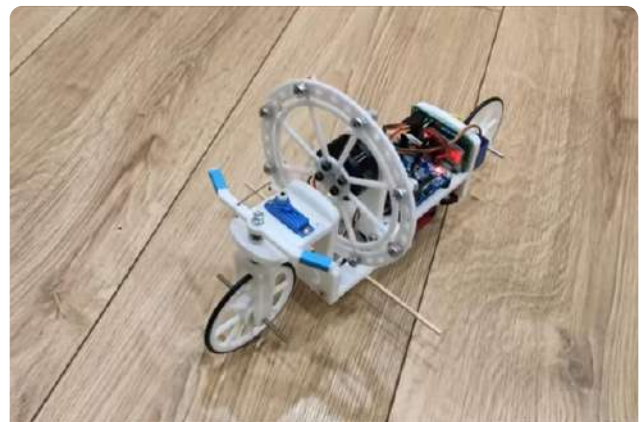
Self Balancing Bot

This project aims to develop a self-balancing robot that utilizes a flywheel mechanism to maintain its equilibrium. The robot is mounted on a mini bike frame, allowing for mobility and dynamic balancing. The main components of the project includes Arduino microcontroller board as the main controller, MPU6050 inertial measurement unit (IMU) for detecting tilt and angular velocity, Servo motor for controlling the flywheel's rotation speed and direction, flywheel mechanism for generating stabilizing torque, mini bike frame with wheels for mobility. The MPU6050 IMU continuously monitors the robot's orientation and angular velocity. The data from the IMU is processed by the Arduino, which then controls the servo motor to adjust the flywheel's rotation speed and direction. The flywheel's angular momentum generates a counter-torque, enabling the robot to maintain its balance on the mini bike frame.

The self-balancing robot on a mini bike demonstrates the principles of rotational dynamics and feedback control systems. The project involves mechanical design considerations for mounting the components securely on the mini bike frame.



A Self Balancing Bot



A Self Balancing Bot





SUMMER PROJECTS

Cleaner Bot

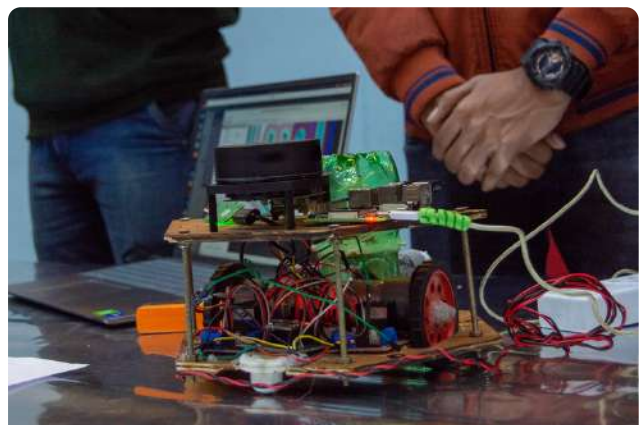
This project aims to develop an autonomous cleaning robot capable of mapping through a residential space using a Raspberry Pi as the primary computing unit and ROS2 (Robot Operating System 2) as the software framework. The robot is equipped with a LiDAR (Light Detection and Ranging) sensor to create a detailed map of the environment and plan efficient cleaning paths.

The key components of the project included a Raspberry Pi single-board computer as the main controller with Arduino Nano as motor controller, ROS2 software framework for robot control and navigation, LiDAR sensor for mapping and localization, Motor drivers and motors for locomotion, Cleaning mechanisms (vacuum cleaner, mop, etc.), Sensor integration (bumper sensors, cliff detectors, etc.), Mapping and navigation algorithms using SLAM Toolbox and Nav2 stack (SLAM, path planning, obstacle avoidance), User interface for monitoring and control, Power management system, Testing and simulation using ROS2 tools (Rviz, Gazebo, etc.)

The robot autonomously navigates through the mapped environment, avoiding obstacles and cleaning the designated areas efficiently. The project leverages the power of ROS2 for modular and scalable development, enabling future enhancements and integrations.



A Cleaner Bot



A Cleaner Bot





SUMMER PROJECTS

Defect Product Detection

This project focuses on creating an automated defect detection system using Convolutional Neural Networks (CNN) to enhance the quality control process in manufacturing. By training the CNN on a diverse dataset of images with various known defects, the model is tuned to identify and classify different types of product imperfections with high accuracy. An integral part of this system is its user-friendly graphical user interface (GUI), developed using Python, which allows even non-technical users to operate the system efficiently. The GUI supports functions such as real-time image analysis and direct image uploads, facilitating immediate defect detection on production lines. This blend of advanced CNN models and an interactive Python GUI aims to significantly reduce defect-related costs and improve production quality by providing swift, accurate defect analysis.

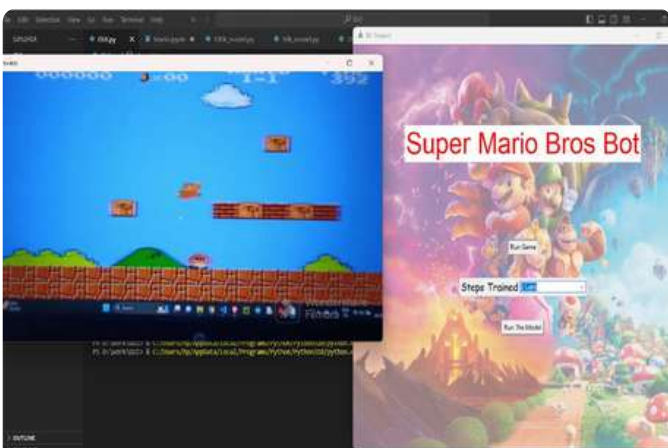


Defect Product Detection

RL in Super Mario Bros

The project aims to leverage reinforcement learning and machine learning to develop a model that can autonomously play the iconic game "Super Mario Bros." This project aims to explore and demonstrate the capabilities of these technologies in understanding and navigating dynamic gaming environments, ultimately showcasing the model's ability to play the game without human intervention.

The main outcome of the project is to make the students better understand reinforcement learning and leverage them according to the use cases.



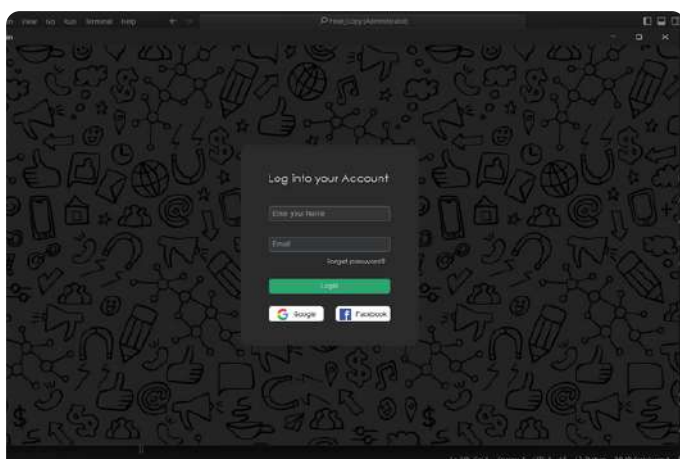
RL in Super Mario Bros



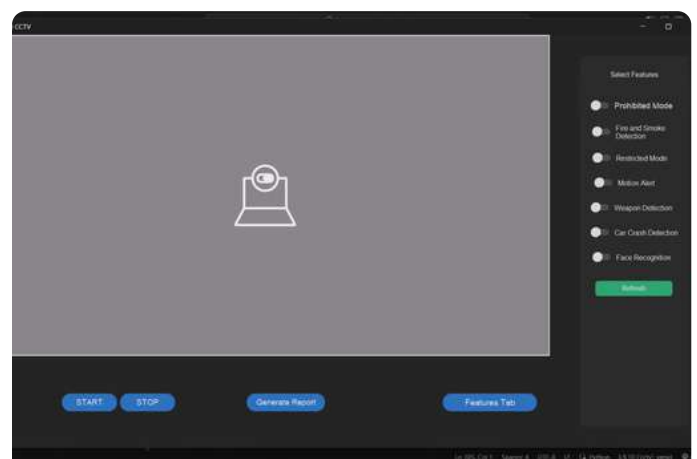
SUMMER PROJECTS

Smart CCTV Surveillance System

This project aimed to develop a Smart CCTV Surveillance System that unlike traditional CCTV Systems not only records and streams videos for surveillance purposes but it can also detect and send an alert email for possible threats like Fire Detection, Smoke Detection, Unwanted Motion Detection, Weapon Detection and can Recognize Face and can generate an Email alert with image and send it to provided Email address. The domain of this project is Computer Vision, it is built using OpenCV and different Python libraries and the GUI of the project is built using Tkinter. This project was equipped with features like Fire and Smoke Detection, Face recognition, Motion Detection, Weapon Detection, Crowd Analytics, Car Accident Detection, Email Alert System, and Report Generation. This Project's outcome was to enhance overall security by providing advanced surveillance features and enabling rapid response to emergencies, accidents, or potential threats.



A Smart CCTV Surveillance System



A Smart CCTV Surveillance System



SELF-DRIVING CAR: A ROBOTICS CLUB ENDEAVOR



Embarking on a journey to revolutionize transportation, the Robotics Club's Self-Driving Car project stands as a beacon of progress. Launched in 2022, the SDC project has swiftly moved from concept to reality, marking a new era of technological advancement.

A Vision Materialized This year, the golf cart, affectionately known as CART 95, underwent a transformative upgrade. The steering system's evolution from a chain drive to a gear drive mesh has significantly enhanced the automated steering functionality, propelling the project into the future.

Engineering Excellence The integration of an absolute single-turn magnetic encoder, coupled with a planetary gear system, has provided critical feedback for steering angles. This breakthrough is instrumental in refining the control system design, setting the stage for the upcoming year's ambitious targets.



BLUEPRINT FOR INNOVATION

Hardware Construction	Component selection and finalizing objectives Prep of 1st budget Prep of high level plan for different modules	Learning Period (Basic Electronics, Development Board)	Complete Electronics and Hardware of Prototype		Building and Deploying Software Stack on Prototype
ROS STACK Skeletons of Node + Visualization Node		Desktop Application for Visualization node	Learning ROS	Construction of ROS Node Skeletons	
Google maps API global planning			Backend support with Google maps API (Direction API)		
Road and Lane Detection		Learning Period (CNN) Understanding previous code and work	Semantic segmentation Model for Road and Lane mask detection in MNNIT		Mapping MNNIT roads in 3-D virtual env.
Lidar based Detection in Carla		Learning Period (Algorithms works in Lidar)	Testing Lidar Algorithms in Carla		Hardware Report on Lidars
SLAM + Odometry + Controllers		Learning Period (Learn standard methods and algorithms)	Test Algorithms & Controllers in Carla		
Obstacle Avoidance and tracking		Learn Local Planners Implement with Webots and other simulators	Test complete DWA in Carla	Explore other Algorithms in SDC	
Reinforcement learning		Exploring Algorithms	Learning Period (David silver Video, Implement RL in Atari Games)		Implement various RL algo in Carla and Deep Racer.
Construction of HD and SD maps + SLAM	Methods to build HD maps in Carla	Building SD maps in Carla & MNNIT and implement global planner	Learning HD Maps	Building HD maps in Carla & MNNIT	

The SDC project is a symphony of complex components and meticulous planning. The chart in the image outlines a comprehensive curriculum and project plan that spans various domains, from **Hardware Construction** to **Learning Periods** and **Retrofit Learning**.

STRUCTURED LEARNING AND DEVELOPMENT



Each category is a stepping stone towards achieving the project's grand vision. The **ROS STACK** serves as the backbone for the project, while components like **Geolocation** and **Visualization Nodes** are crucial for the SDC's spatial awareness.

Tasks such as "**Solutions MELs API visual programming node**" and "**Prep of high-level plan for retrofit modules**" highlight the collaborative nature of the project, where every member contributes to the collective goal.



COLLABORATIVE EFFORTS





THE FIRST MILESTONE: SDC'S INITIAL PROTOTYPE

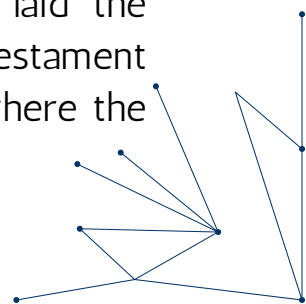


Testing of Local Navigation on SDC prototype

Under the cloak of night, the first prototype of the SDC project took to the streets, its green lights a beacon of the progress to come. This small four-wheeled vehicle, captured in the image, was more than just a machine; it was a vision brought to life.

A Test of Potential The initial prototype served as a crucial testbed for the technologies that would later be refined in CART 95. It was here, on the open road, that the early iterations of navigation and control systems were put through their paces.

Illuminating the Path Forward The success of this prototype laid the groundwork for the advancements that followed. It stands as a testament to the ingenuity and determination of the team, a reminder of where the journey began and the limitless potential that lies ahead.





The Robotics Club's first self-driving car successfully navigates MNNIT's campus, marking a milestone in autonomous vehicle innovation.

A JOURNEY TOWARDS SUSTAINABLE MOBILITY

As we conclude our exploration of the Self-Driving Car (SDC) project, we are reminded of the strides taken towards a sustainable and autonomous future. The image of the electric golf cart, nestled within the green embrace of nature, symbolizes the harmony between technological advancement and environmental stewardship.

The Promise of Tomorrow The SDC project, with its innovative spirit and forward-thinking approach, stands at the forefront of this movement. It is a beacon of hope, illuminating the path towards a world where transportation is not only smart but also kind to our planet.

A Legacy of Innovation As we look ahead, the legacy of the SDC project will be defined by its contributions to the field of autonomous vehicles. The journey does not end here; it is merely the beginning of a new chapter in the story of mobility.

The Horizon Awaits The future beckons with endless possibilities, and the SDC project is poised to meet it head-on. With each passing year, we inch closer to a reality where self-driven cars are an everyday sight, transforming the way we live, work, and connect.



BEHIND THE DEVELOPMENT OF SELF-DRIVING VEHICLE

A MECHANICAL ENGINEERING DEPARTMENT & ROBOTICS STUDENT CLUB INITIATIVE

SPONSORED & INITIATED BY

MNNIT Alumni Excellence (MAE) Foundation
A 1995 BATCH INITIATIVE

Under Technical Guidance of Professors

“

Innovation and creativity are at the core of robotics, constantly pushing us to explore new frontiers.

”



Dr. Samir Saraswati
Associate Professor (MED)
Faculty In-charge
& Mentor -
Self Driving Car



Dr. Jitendra Narayan
Gangwar
Assistant Professor
Grade - 1
Faculty In-charge & Mentor
- Self Driving Car

“

Robotics isn't about machines; it's about creating companions that enhance human capabilities.

”



Rohit Garg
- Head member of
MNNIT Alumni Excellence

“

Establishing prototyping as a core competence for innovation, requires more than creating a research and development department or team. It requires school wide value for innovation.

”

Under Technical Mentorship of Alumni

Bhuvan Jhamb Alumni, Batch-2020 R&D Engineer, Intel, India	Kishan Tiwari Alumni, Batch-2019 Founder, TSAW Drones, India	Sharad Rawat Alumni, Batch-2016 Autonomous Driving Software Engineer, Germany
---	---	--

The project was proceeded by
following SDC Heads :

Passing Year	Name
2020	Gaurav Bansal
2021	Bhuvan jhamb
2022	Ashutosh Kumar
2023	Amit Gupta
2024	Ayush Singh Gour
2025	Rishi Mishra





GLOBAL ALUMNI CONVENTION

The Robotics Club's participation in the Global Alumni Meet (GAC), organized by the MNNIT Alumni Association (MAA), brings immense benefits to both alumni engagement and technological innovation. By showcasing a diverse range of projects, including hardware and software innovations, the club highlights the talent and expertise of alumni while inspiring current students and fellow alumni to explore cutting-edge technologies.



Through the GAC platform, the Robotics Club facilitates meaningful interactions among alumni, fostering collaboration and knowledge exchange. Alumni have the opportunity to network with industry leaders, share insights, and explore potential partnerships or projects. This engagement strengthens the alumni community and opens doors for collaborative initiatives that can drive advancements in various fields.

Moreover, the Robotics Club's presence at GAC enhances the event's appeal and relevance, attracting a wide audience interested in emerging technologies. The showcase of projects like humanoid robots, robotic arms, and self-balancing bots captivates attendees and demonstrates continuous innovation within the alumni network.

एमएनएनआईटी के छात्रों ने तैयार किया चैटबाट

जागरण विशेष
बृहस्पति, 14 अक्टूबर

भारतीय नैतिकता के संरक्षण (एमएनएनआईटी) के छात्रों ने चैट-बोट को एक नैतिक लेखन प्रोग्राम के माध्यम से प्रस्तुत किया है। आर्टिफिशियल इंटेलिजेंस के माध्यम से एमएनएनआईटी के छात्रों को गतिविधियों के बारे में जानकारी प्रदान करने में मदद करता है। चैटबोट के सर्वर को जोड़ने के बाद छात्रों को आर्टिफिशियल इंटेलिजेंस का प्रयोग करने से यह चैटबोट चलाया जा सकता है। चैटबोट के सर्वर को जोड़ने के बाद छात्रों को आर्टिफिशियल इंटेलिजेंस का प्रयोग करने से यह चैटबोट चलाया जा सकता है।



भारतीय नैतिकता के संरक्षण (एमएनएनआईटी) के छात्रों ने चैट-बोट को एक नैतिक लेखन प्रोग्राम के माध्यम से प्रस्तुत किया है। आर्टिफिशियल इंटेलिजेंस के माध्यम से एमएनएनआईटी के छात्रों को गतिविधियों के बारे में जानकारी प्रदान करने में मदद करता है। चैटबोट के सर्वर को जोड़ने के बाद छात्रों को आर्टिफिशियल इंटेलिजेंस का प्रयोग करने से यह चैटबोट चलाया जा सकता है।

को अभी एमएनएनआईटी के छात्रों के विभिन्न कक्षाओं को गतिविधियों के माध्यम से ही प्रस्तुत किया गया है। ऐसे में अभी इसका प्रयोग संभव नहीं हो रहा है पर जितने छात्रों से इसकी प्रतिक्रिया मिलेगी वह उतना ही लाभकारी होगा।

चैटबोट से रीसर्च जल्दी करवा है चैटबोट : छात्रों ने बताया कि चैटबोट रास का उपयोग करते हुए आर्टिफिशियल इंटेलिजेंस के लिए एक भाषा प्रसंस्करण सफ्टवेयर एप्लिकेशन को एकीकृत किया गया है।

यह प्रश्नों का उत्तर दे सके इसके लिए एक डेटासेट तैयार किया जाता है और और उपलब्ध भाषा प्रसंस्करण उपकरण का उपयोग करके अपने डेटासेट के लिए इस माध्यम को प्रशिक्षित करते हैं इससे हर ट्रेनिंग यानी चैटबोट से सीखने के सवा-सवा प्रारंभिक जानकारी को वेबो से प्राप्त करने इसकी क्षमता में यह डेटासेट के प्रशिक्षण से निरंतर सुधार होता होगा।

पहली स्वदेशी मानवरहित कार ने पार की दूसरी बाधा

एमएनएनआईटी परिसर की सड़कों पर 30 किमी प्रति घंटा की रफ्तार से चली, अवरोधक दिखने पर रुकी और मुड़ भी गई

अमर उजाला ख्युरो

प्रयागराज। देश की पहली मानवरहित कार का बुधवार को एमएनएनआईटी परिसर की सड़कों पर दूसरे चरण का परीक्षण किया गया, जो सफल रहा। परिसर की सड़कों पर 30 किमी प्रति घंटा की रफ्तार से दो सौ मीटर से अधिक दूरी इस कार ने तय की। इस दौरान मानवरहित कार अवरोधक मिलने पर न सिर्फ रुकी, बल्कि अपने आप ब्रेक लेने और छोड़ने के साथ ही रास्ता भी बदलती रही। विद्यार्थी, चौराहों के टर्निंग प्वाइंट पर खुद मुड़कर यह कार अपने निर्धारित स्थल पर पहुंच गई। हालांकि, इस परियोजना को फिनिशिंग लाइन तक पहुंचाने के लिए अभी कुछ काम बाकी रह गया है। वहीं, उम्मीद जताई जा रही है कि अगले छह महीने में यह



एमएनएनआईटी परिसर में मानवरहित कार का परीक्षण किया गया। संवाद

मानवरहित कार सड़कों पर दौड़ती नजर आ सकती है।

पांच सैटी वाली मानवरहित कार के दूसरे चरण के सफल परीक्षण को लेकर एमएनएनआईटी के टेक्नोक्रैट श्रेय उल्लेखित हैं। वर्ष भर पहले यह कार इसी परिसर में सिर्फ 30 मीटर ही चल सकी थी। उस समय यह कार स्टोयोरिंग चैन सिस्टम पर आधारित

थी। कोडिंग और प्रोग्रामिंग के बाद अब अमेरिका की फर्निगी मेल्विन यूनिवर्सिटी के कंप्यूटर विज्ञान विभाग के शोध छात्र भुवन जाम को मदद से इसकी अन्य बाधाओं को पार कर लिया गया है।

भुवन की मदद से मैकेनिकल डिजाइन, इलेक्ट्रिकल, कंप्यूटर कोडिंग और विज्ञान पर इस परियोजना से जुड़े



अमेरिका की फर्निगी मेल्विन यूनिवर्सिटी की मदद से मानवरहित कार को स्टोयोरिंग और विज्ञान की बाधाओं को दूर करने में सफलता मिल गई है। इस कार को अब जहाँ भी जाने के लिए कोडिंग के माध्यम से कहा जाएगा, वहाँ वह अपने आप सस्ता दूढ़ते हुए पहुंच जाएगी। - आर्युष गौर, परियोजना के सिविल चर्क हेड

19 छात्रों की टीम ने काम किया। कार की स्टोयोरिंग को अब चैन रहित बनाने में सफलता मिल गई है। इस टीम के मैकेनिकल हेड आर्युष गौर हैं। इलेक्ट्रिकल के प्रभारी शशांक सिंह और कंप्यूटर विज्ञान के हेड अनुराग हैं। पूरा छात्र सम्मेलन के दूसरे और आखिरी दिन दोपहर बाद एमएनएनआईटी के कन्वेंशन

भवन परिसर से निकल कर दो सौ मीटर तक फर्श भरने के बाद इस मानवरहित कार को पूरी तरह सड़कों पर चलने के लिए फिट पाया गया।

अनुराग गौर ने बताया कि इस कार का अब तीसरा और आखिरी चरण शरभ होगा, जो छह महीने चलेगा। एमएनएनआईटी के प्रोफेसर समीर और प्रो. जितेंद्र गंगवार के निदेशन में बीटेक छात्रों की टीम ने मिलकर इस कार को तैयार किया है। इस कार के लिए फंडिंग और मार्गदर्शन एमएनएनआईटी के ही साल 1995 बैच के छात्र रहे अरुणराजी गो क्लिएटिव के क्लस प्रेसिडेंट रोहित गर्ग और सड़क एवं परिवहन मंत्रालय नई दिल्ली में अधीक्षण अभियंता के पद पर तैनात रानेश कुमार ने किया है। कार्ट-95 के नाम से इस मानवरहित कार परियोजना पर 15 लाख रुपये लागत आई है।

The Robotics Club successfully tested their first indigenous self-driving car on MNNIT's campus, navigating obstacles at a speed of 30 km/h, marking a significant milestone in autonomous vehicle technology.

The Robotics Club's involvement in GAC not only showcases technological prowess but also serves as a platform for alumni to reconnect with their alma mater. These interactions often lead to valuable mentorship opportunities, where experienced alumni guide current students and recent graduates, helping them navigate their career paths and entrepreneurial ventures.

Additionally, the club's diverse project presentations at GAC offer a glimpse into the latest trends and advancements in robotics and technology. Attendees can witness firsthand the practical applications of theoretical knowledge, bridging the gap between academia and industry. This exposure is crucial for current students, who gain insights into real-world challenges and solutions.

The Global Alumni Meet, with the Robotics Club's active participation, becomes a melting pot of ideas and innovations. It not only reinforces the institution's commitment to fostering a culture of continuous learning and innovation but also strengthens the bonds within the alumni community. By contributing to the event's success, the Robotics Club plays a pivotal role in promoting technological advancements and inspiring the next generation of innovators.



The Robotics Club's stall at the Global Alumni Meet showcases innovative projects like humanoid robots, robotic arms, and self-balancing bots, highlighting the club's expertise and advancements in technology.



2023-24



MOMENTS ADMIST EXCELLENCE

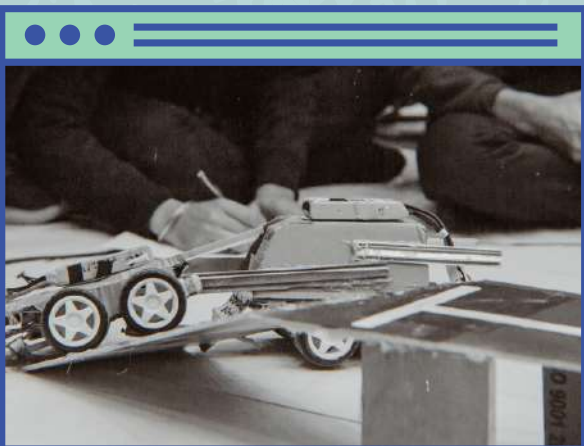
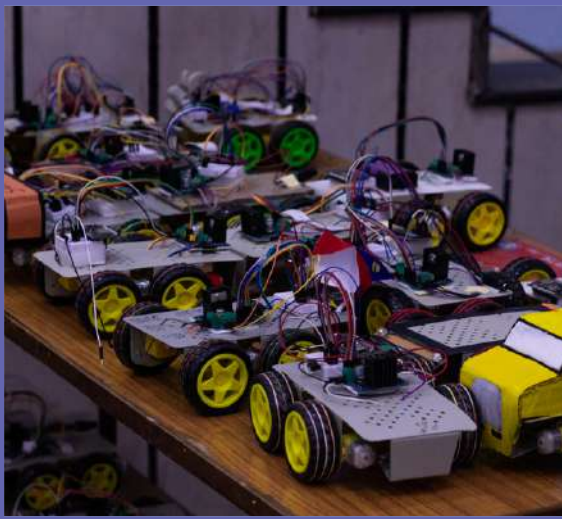
E V E N T S





HANDS-ON INNOVATION :

Annual Robotics Club Events at MNNIT Allahabad



The Robotics Club at MNNIT Allahabad organizes several events each year, providing a platform for technocrats to showcase their skills and gain hands-on experience through various competitions. Prominent events include BotRush and Robomania - Avishkar, which foster innovation and practical learning in the field of Robotics. These events are meticulously designed to challenge participants and encourage creative problem-solving. Through teamwork and collaboration, students not only enhance their technical proficiency but also develop essential soft skills. The Robotics Club's initiatives aim to inspire a passion for robotics and technology, preparing students for future advancements and career opportunities in these dynamic fields.

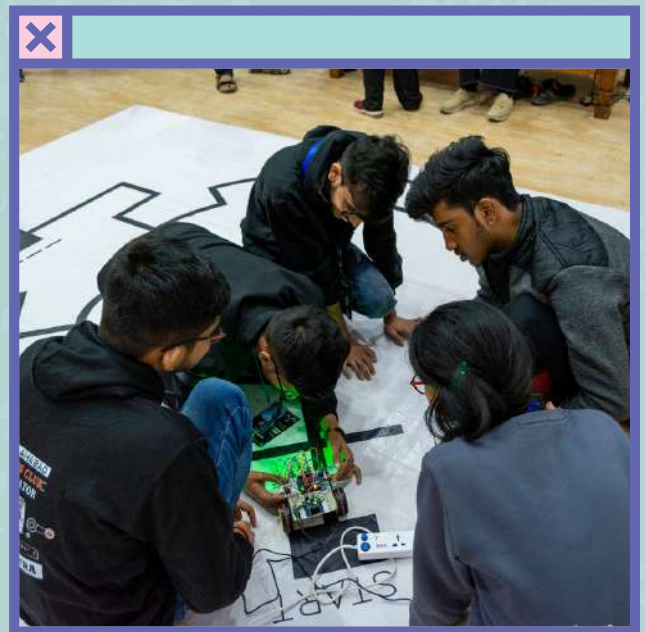




BOTRUSH

BotRush stands out as a flagship annual robotics event tailored specifically for first-year students at MNNIT Allahabad. It serves as a crucial stepping stone for newcomers, offering them a unique opportunity to dive into the world of robotics in a supportive and engaging environment.

With a focus on introductory challenges and beginner-friendly competitions, BotRush provides a platform for budding technocrats to explore their interests, refine their skills, and forge lasting connections within the robotics community. This event not only ignites a passion for innovation but also lays the foundation for future success in the field of robotics.





EVENTS UNDER BOTRUSH

Force Secure

Picture an advanced access system requiring a unique password for entry, ensuring only authorized individuals can open the door. It also includes a presence detection feature to monitor personnel smoothly. The interface is user-friendly, designed with holographic technology for secure access and real-time monitoring. Additionally, an alert system is in place to detect and respond to unauthorized attempts, activating surveillance equipment for added security.



Cell Phone Controlled Robot

Teams of engineers are embarking on a project to develop cell phone-controlled robots with sophisticated capabilities. These robots are designed to navigate through a challenging arena filled with various obstacles and trials. Their mission includes maneuvering through hazardous pathways, evading simulated threats, and adjusting to dynamic terrains. The goal is for these robots to demonstrate not only agility and resilience but also strategic thinking and problem-solving abilities as they tackle the tasks presented in the trials.



Line follower Robot

The challenge invites engineers and robotics enthusiasts to create and program their robots with advanced sensors. These robots must navigate through dark paths while avoiding dangers. As they move through the maze, they encounter holographic illusions presenting various challenges, testing their problem-solving skills and ability to maneuver accurately.

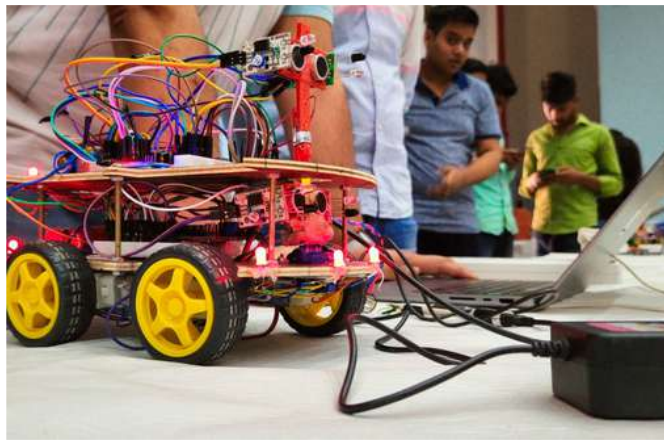




EVENTS UNDER BOTRUSH

Force Innovate

Your mission is straightforward: develop droids capable of outsmarting cunning adversaries. Consider incorporating advanced technology, intuitive sensors, and intelligent AI. Imagine seamlessly merging cutting-edge tech with innovative concepts. This challenge isn't just an opportunity; it's an invitation to push the boundaries of innovation. Unleash your creativity and invent droids that redefine conventional norms. Your creations could pave the way for a new era of advancement!



Botwars

Your mission is clear: design bots that outwit even the most ingenious adversaries. Think advanced features, intuitive sensors, and intelligent algorithms. Imagine seamlessly blending state-of-the-art technology with innovative concepts. This challenge isn't just an opportunity; it's a call to push the boundaries of innovation. Unleash your creativity and invent bots that redefine conventional norms. Your creations could pave the way for a new era of advancement!





TECH-EXPO

The Botruh 2.0 expo showcased the pinnacle of student innovation in robotics. The event was graced by the esteemed presence of our Director and the Student Activity Center (SAC) President, who both commended the students' inventive efforts.

Innovation Alley was abuzz with activity, featuring stalls that displayed a variety of robots, including both software and hardware model. Each stall offered a glimpse into the future of robotics, with projects ranging from simple mechanical constructs to complex autonomous systems.

The expo was not just a display of technological prowess but also a platform for dialogue, with the Director, SAC President and club's faculty in-charge engaging with students, fostering a spirit of collaboration and learning.

As we reflect on the success of Botruh 2.0, we are reminded of the power of community and innovation in shaping the future of technology.





Workshops



Workshops organized by the robotics club on various aspects of robotics, including programming, electronics, mechanics, and AI.

The club regularly organizes workshops and doubt sessions covering a wide array of topics essential to robotics. These sessions include programming, electronics, which delves into circuit design and hardware integration; mechanics, focusing on the physical construction and movement of robots; and artificial intelligence (AI), exploring machine learning and intelligent behavior in robotic systems. Through these activities, the club aims to equip students with the practical skills and theoretical knowledge necessary to excel in the rapidly evolving field of robotics, encouraging innovation and collaboration.

THE MINDS BEHIND.





ROBOMANIA - AVISHKAR

Robomania is a prominent event that takes place during Avishkar, the annual tech fest of Motilal Nehru National Institute of Technology (MNNIT) Allahabad. This event is one of the highlights of the fest, attracting participants and enthusiasts from various technical and engineering backgrounds.

Robomania is a dedicated segment of Avishkar that focuses on robotics. It includes a series of competitions and challenges designed to test the skills of participants in designing, building, and programming robots.



Now what is Avishkar?

Avishkar is MNNIT's premier tech festival, showcasing a blend of innovation, technology, and creativity. The fest usually spans several days and includes a variety of events such as workshops, competitions, guest lectures, and exhibitions. Avishkar provides a platform for students to demonstrate their technical prowess, engage in competitive activities, and collaborate on innovative projects.





EVENTS UNDER ROBOMANIA

Cozmo-Clench

Cozmo-Clench challenges participants to construct a wirelessly controlled robot that can grip various objects and place them in designated target zones. The robot must have a reliable gripping mechanism for handling objects of different shapes and sizes. Participants develop a precise wireless control system to maneuver the robot remotely, ensuring accurate placement of objects. Key challenges include achieving stable wireless connectivity, designing an effective gripper, and navigating the robot precisely to place objects correctly.



Hack Quest

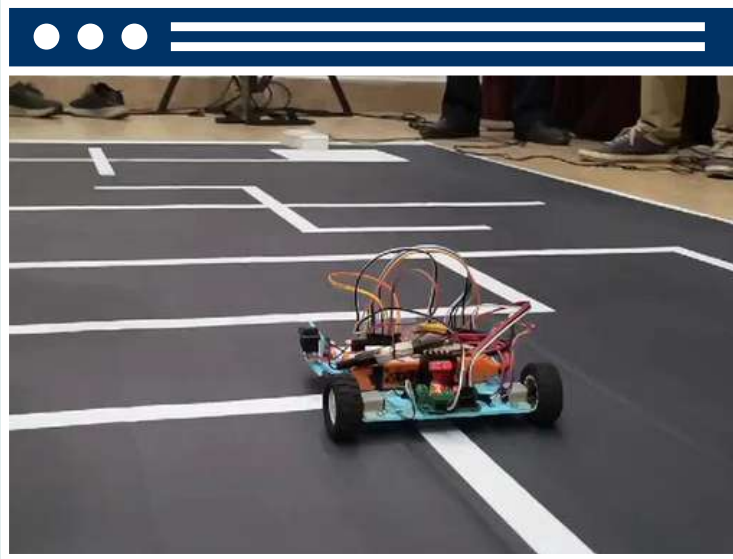
Hack Quest is an idea presentation event where participants propose innovative robotics projects and solve problems using machine learning. Participants conceptualize impactful projects and effectively utilize machine learning techniques to address challenges or improve systems. This event emphasizes innovative thinking, problem-solving, and effective communication, requiring participants to present their ideas clearly and convincingly.





EVENTS UNDER ROBOMANIA

Track-n-Trek



Track Trek focuses on building a line-following robot that navigates a track using the shortest path algorithm. The robot needs sensors to detect and follow a line consistently. It must calculate and follow the shortest path to complete the track quickly. This event challenges participants to implement robust sensor integration, handle complex track layouts, and optimize pathfinding algorithms, honing skills in real-time navigation and algorithm development.

Navigate the Dark



Navigate The Dark involves constructing a wirelessly controlled robot that solves a hidden maze using a live camera feed. The robot must have cameras for real-time video feedback and an algorithm to interpret the feed for navigation. This event tests real-time image processing, algorithm development, and low-visibility navigation. Challenges include integrating the camera, processing live feeds for navigation, and ensuring accurate maze solving based on camera input.





Detect N Direct

Detect N Direct requires building an autonomous device that detects target circles and accurately points a laser at their centers. The device uses image recognition to locate targets and a precise mechanism to aim the laser. This event challenges participants to develop robust image recognition algorithms and precise control systems, ensuring the device autonomously locates and targets circles accurately. Skills in image processing and autonomous control are crucial for success.



2023-24

ACHIEVEMENTS

“

Science is a way of thinking much more
than it is a body of knowledge.

-Carl Sagan

”



E-YANTRA {IIT BOMBAY}

Our robotics club has demonstrated exceptional prowess and innovation at eYantra, the prestigious robotics competition hosted by IIT Bombay. This year, our dedicated team showcased their technical skills and problem-solving abilities, securing remarkable achievements that reflect our commitment to excellence and our passion for robotics.



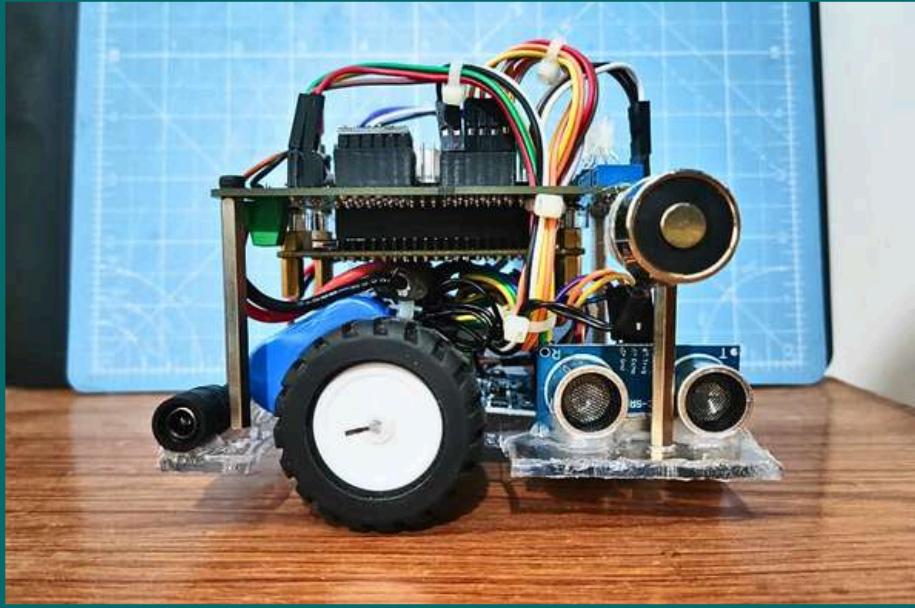
The Robotics Club won the second Prize in the E-Yantra Robotics Competition under the theme Astro Tinker Bot – An FPGA-based Robotic System for Autonomous Path Planning and Navigation.

The "Astrotinker Bot" project aimed to develop an advanced robotic system capable of calculating the shortest path between two point and autonomously navigating that path. The project leveraged the power of Field-Programmable Gate Arrays (FPGAs) and custom-designed hardware to achieve this goal.

“Technology, much like art, represents a soaring exercise of the human imagination.”

”





PROJECT FEATURES

1. Custom RISC-V CPU Implementation:

At the core of the project was a custom-designed RISC-V CPU, implemented on an FPGA. This CPU was responsible for executing complex path-planning algorithms and controlling the robot's navigation.

2. Path Planning Algorithms:

The team developed and optimized algorithms to calculate the shortest path between two given points, taking into account obstacles and other environmental factors.

3. Autonomous Navigation:

Once the shortest path was determined, the robot was programmed to autonomously navigate through the designated route, avoiding obstacles and adjusting its trajectory as needed.

4. Sensor Integration:

The robot was equipped with various sensors, such as ultrasonic or infrared sensors, to detect obstacles and gather environmental data for accurate path planning and navigation.

5. FPGA-based Hardware Design:

The team designed and implemented the hardware components, including the FPGA board, motor controllers, and sensor interfaces, to integrate seamlessly with the custom RISC-V CPU and software components.





“The journey in the competition has been nothing short of remarkable. Each member of our team contributed their unique skills and expertise, pushing boundaries and overcoming challenges along the way that have kept us in the Top 3 throughout the competition.”

THEME ASTROTINKER BOT 2ND PRIZE WINNERS

Team Members:

Sudhanshu Ranjan (20215049)

Sumit Saroj (20215155)

Md Faizan (20212027)



The Robotics Team at IIT Bomabay

TECHFEST IIT BOMBAY



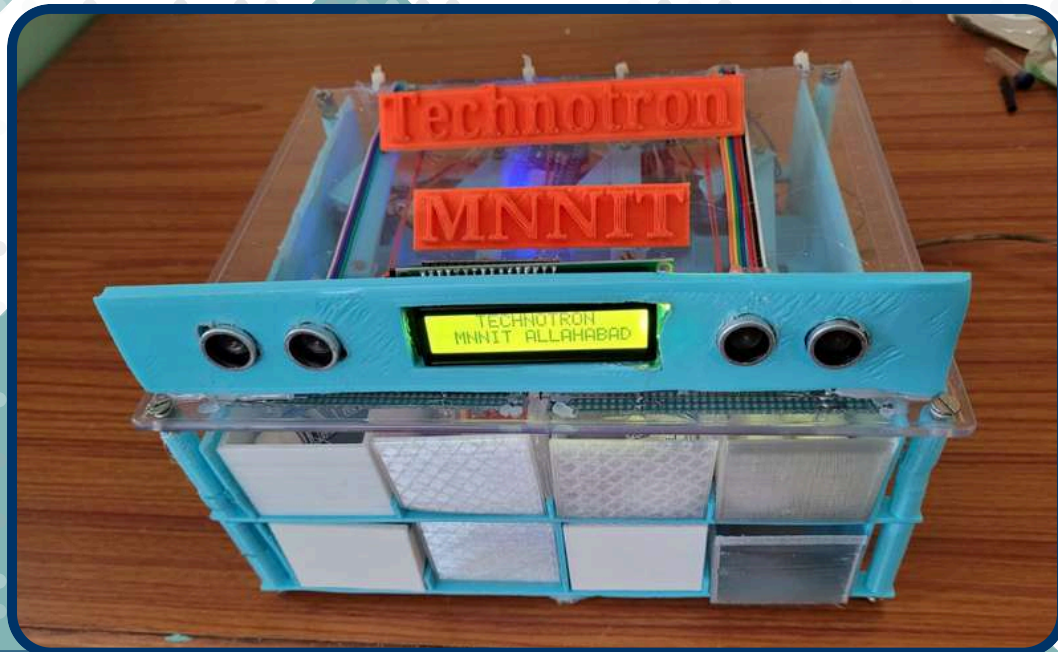
As the year came to an end the Robotics Club participated in the prestigious TechFest held at IIT Bombay where students faced great competition from all over the world in the field of Robotics, artificial intelligence, and machine learning.

Among them, our stood out, leaving an indelible mark with their exceptional performance. Throughout the tech fest, we showcased our expertise and determination, emerging as finalists in events such as Meshmerize, Cozmodench, Tech Aid, Hack-Ai, and many more securing commendable positions. However, it was in the thrilling AtomQuest event that we truly shined, ultimately clinching the second prize, solidifying our position as formidable contenders in the realm of robotics.

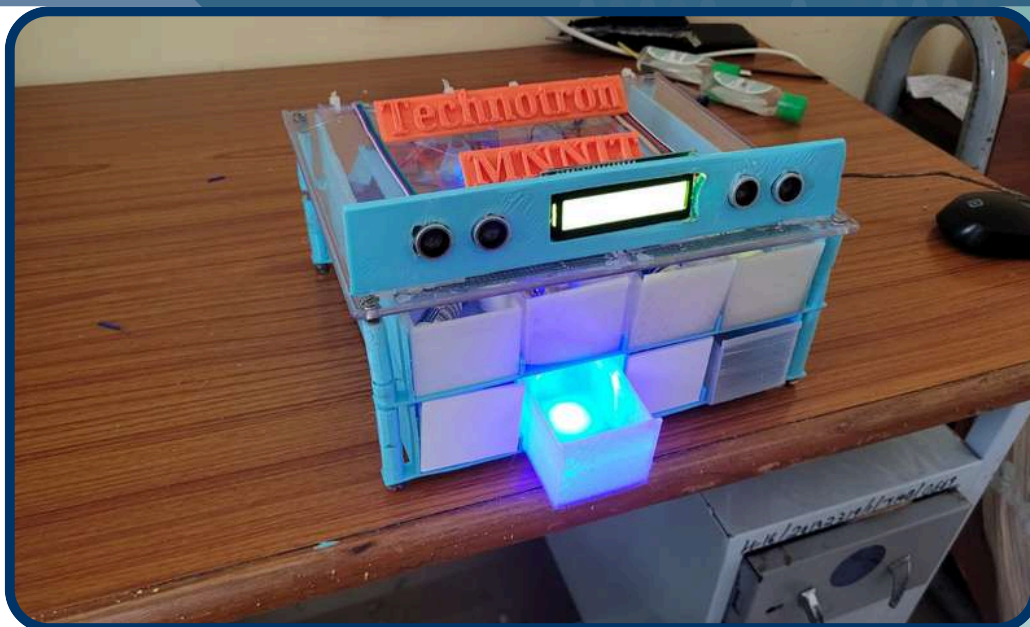




VOICEVAULT {VOICE CONTROLLED SMART DRAWER SYSTEM}



Our AtomQuest Team (Team Technotron) had created a voice-based smart drawer system that opens and closes drawers using voice commands according to the objects present inside them. This can be a convenient and innovative addition to your home, providing hands-free access to drawer contents and reducing the time it takes to search for things we need.





PROJECT BENEFITS

1. CONVENIENCE AND HANDS-FREE OPERATION:

- Hands-free access
- Convenient in busy or dirty situations
- Eliminates the need for physical interaction

2. ENHANCED ACCESSIBILITY FOR INDIVIDUALS WITH DISABILITIES:

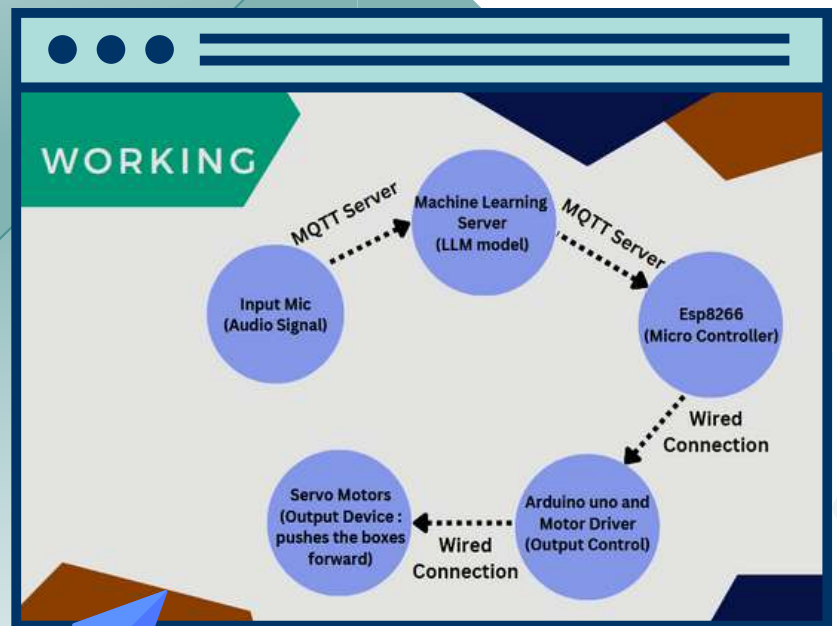
- Inclusivity, independence through an alternative to manual handling
- Improved accessibility for individuals with physical disabilities.

3. CUSTOMIZATION AND PERSONALIZATION:

- Customizable item naming and categorization
- Easy retrieval with simple vocal commands
- Enhanced user experience through customization.

4. EFFICIENCY IN EVERYDAY RETRIEVAL:

- Quick and precise item finding method
- Verbal requests replace manual searching
- Substantial reduction in time and effort for daily retrieval
- Streamlined process adds convenience to daily tasks
- Improved overall user-friendliness





A BEACON OF EXCELLENCE

*Securing the **Second Prize** in the AtomQuest event was a moment of pride for our club and a testament to the collective effort and dedication of each member. It was not merely a victory but a reaffirmation of our commitment to excellence and our ability to push the boundaries of what is possible in the field of robotics.*

Members:

1. Sarthak Kumar (Leader) [3rd Year]
2. Bipul Karna [4th Year]
3. Avneesh Sahu [2nd Year]
4. Piyush Chourasia [2nd Year]

**TEAM - TECHNOTRON
2ND PRIZE WINNERS**



*Our Atomquest Team
(Atom-230249)*

“

We commend our AtomQuest squad for their dedication, innovation, and exceptional performance, earning them the well-deserved second prize at the AtomQuest event. Their dedication, ingenuity, and tireless effort have truly shone through, showcasing not only their immense talent but also their ability to thrive in challenging environments.

”





LOOKING AHEAD

As we reflect on our triumph at IIT Bombay TechFest, we are filled with gratitude for the support of our mentors, the enthusiasm of our fellow club members, and the unwavering encouragement of our college community. With this victory as inspiration, we now set our sights on new horizons, eager to continue our journey of exploration, innovation, and impact in the world of robotics.

Curious about our club? Wondering what we stand for and what activities we offer? Interested in how you can become a member and join our vibrant community? Visit our official website to explore all the details about our mission, events, membership benefits, and more. We can't wait to welcome you!

<https://sac.mnnit.ac.in/robotics/>





EXPANDING OUR HORIZONS



Participating in the prestigious TechFest at IIT Bombay was not only a testament to our club's dedication but also an opportunity to expand our reach and network within the global robotics community. Engaging with students and experts from around the world, we exchanged ideas, forged new connections, and gained invaluable insights into the latest advancements in robotics, artificial intelligence, and machine learning.





CHALLENGES AND TRIUMPHS

The journey to success at TechFest was not without its challenges. From countless hours of rigorous preparation to navigating through intense competition, every step of the way tested our resilience and determination. Yet, fueled by our passion for innovation and collaboration, we rose to the occasion, overcoming obstacles and triumphing against the odds.

“

“Arise! Awake! and stop not until the goal is reached”

-Swami Vivekanand

”



BEHIND THE CLUB



*Prof. Rama Shanker Verma
- Director*

“

The first step in teaching students to innovate is making sure that educators have opportunities to be innovators themselves.

”



*Dr. Pritam Singh
-SAC President*

“

Technology, much like art, represents a soaring exercise of the human imagination.

”

“

Robotics is the pinnacle of superb dynamics and precise position control in this era of automation.

”



*Dr. Anand Sharma
-Faculty Coordinator
Robotics Club*



*Prof. Asim Mukharjee
-Faculty In-charge
Technical Activities*

“

With unwavering willpower, the Robotics Club at MNNIT is gearing up for the battles in the dynamic world of robotics.

”

SPECIAL MENTIONS

MNNIT Alumni Association (MAA)



“

Imagination is not only the uniquely human capacity to envision that which is not, and therefore the fount of all invention and innovation.

In its arguably most transformative and revelatory capacity, it is the power to that enables us to empathize with humans whose experiences we have never shared.

”

&

MNNIT Alumni Excellence (MAE) Foundation

“

Driven by innovation and creativity, we transform visionary ideas into groundbreaking technological realities.

”



Club Coordinators '24



Shashank Singh



Bipul Karna



Gautam Kumar



Ayush Gaur



Ankit



Abhishek Hanotiya



Prakhar Agrawal



Shreya Gupta



Aditya Jain

Video Team



Prince Yadav

Design Team



Ishika Agrawal

Club Coordinators '25



Aayush Verma



Rishi Mishra



Sudhanshu Ranjan



Sarthak Kumar



Devendra Saini



Ashay Jadhav



Anu Priya



Bhanu Pratap Singh



Amisha Singh



Tushar Kesarwani



Non Tech Members



Devanshi Gupta
Designer



Kavya Gupta
Designer



Abhinav Anand
Designer



Vatsal Kumar
Designer



Kanishk Agrawal
Designer



Shourya Mishra
Designer



Ankit Kumar
Designer



Dhruv Chandhok
Video Editor



Isha Jaiswal
Video Editor

"Robotics and automation will be the drivers of the next industrial revolution, shaping the future of technology and society."

– John Doe



Keep in touch.

Visit <https://sac.mnnit.ac.in/robotics/> or follow us on social media to learn about all that is happening at Robotics Club MNNIT



@roboticsclubmnnit



Designed by:
Robotics Design Team